

Autonomous Object Tracking Car

YOLOv8 + Arduino Serial Communication Guide

Project Overview

This project creates a closed-loop control system where a Python-based Vision Brain (running YOLOv8) communicates with an Arduino-based Mechanical Driver. The car identifies a target object, calculates its position relative to the camera frame, and adjusts its motors to chase the target.

1. The Python "Brain" (Computer Vision)

The Python script is responsible for three main tasks: Image Acquisition, Object Detection, and Command Generation.

Key Logic: The Error Calculation

The core of the navigation is calculating the horizontal "offset". If the image width is W , the center is $C = W / 2$. If the detected object center is O_x , the offset is:

$$Error = O_x - C$$

The code uses a "Deadzone" strategy. If the error is small (e.g., between -60 and 60), the car moves straight to prevent jittering.

Python Code Breakdown

```
# Loading the Model and Serial
model = YOLO('/content/drive/MyDrive/test/runs/train_data/weights/best.pt')
arduino = serial.Serial('COM3', 9600)

# Main Loop
while True:
    results = model(frame)
    if results[0].boxes:
        # Calculate Object Center
        x1, y1, x2, y2 = box.xyxy[0]
```

```

obj_center = (x1 + x2) / 2

# Determine Direction
if obj_center < (img_center - 60):
    command = "L150" # Turn Left
elif obj_center > (img_center + 60):
    command = "R150" # Turn Right
else:
    command = "F150" # Forward
else:
    command = "T120" # Search Mode (Rotate)

arduino.write(command.encode())

```

2. The Arduino "Muscles" (Hardware Control)

The Arduino receives strings like "L150". It separates the character (Direction) from the number (PWM Speed).

Pulse Width Modulation (PWM)

We use `analogWrite(pin, speed)` to control the velocity. 0 is fully off, 255 is fully on. This allows the Python script to dynamically slow down or speed up the car based on the user's configuration.

Arduino Command Parsing

Command Prefix	Action	Motor State (L/R)
F	Forward	Both Motors Forward
L	Left Turn	Left Backwards / Right Forward
R	Right Turn	Left Forward / Right Backwards
T	Rotate Search	Clockwise Spin

Critical Implementation Notes

- **Baud Rate:** Both the Python script and `Serial.begin()` must be set to 9600.
- **Power Supply:** Use a separate 7.4V or 11.1V battery for motors; do not power them via the Arduino's 5V pin.
- **Paths:** For Google Colab, ensure your model is at `/content/drive/MyDrive/...` and your local laptop script points to the correct USB COM port.